

1. The $ETS(AAdN)$ model is given by the system

$$\begin{cases} u_t \sim \mathcal{N}(0; 100) \\ b_t = 0.4b_{t-1} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.4b_{t-1} + 0.1u_t \\ y_t = \ell_{t-1} + 0.4b_{t-1} + u_t \end{cases}$$

with $\ell_{100} = 30$ and $b_{100} = 2$.

- (a) [5] Find the 95% predictive interval for y_{102} .
 - (b) [5] Find conditional probability $\mathbb{P}(y_{102} > 40 \mid \ell_{100}, b_{100})$ in terms of the standard normal distribution function.
2. The semi-annual y_t is modelled by $ETS(ANA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 16) \\ s_t = s_{t-2} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.5u_t \\ y_t = \ell_{t-1} + s_{t-2} + u_t \end{cases}$$

with $\ell_0 = 20$, $s_0 = 5$ and $s_{-1} = -5$.

- (a) [4] Find best point forecasts for y_{100} and y_{101} .
 - (b) [6] Assume the first observation y_1 is missing in the dataset, but the second one is known, $y_2 = 30$. Find the best point forecast for y_1 .
3. Consider $MA(1)$ process $y_t = 10 + u_t + 5u_{t-1}$, where u_t is a white noise process with variance 20.
- (a) [5] Find $\mathbb{E}(y_t)$, $\text{Var}(y_t)$, $\text{Cov}(y_t, y_s)$.
 - (b) [1] Is the process (y_t) stationary?
 - (c) [1] Is the process (y_t) invertible with respect to (u_t) ?
 - (d) [3] If possible represent y_t in the form $y_t = \alpha + v_t + \beta v_{t-1}$ where α and β are some constant with $\beta \neq 5$ and (v_t) is a white noise.
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4. Consider the equation $y_t = 1 + 1.2y_{t-1} - 0.5y_{t-2} + u_t - 0.8u_{t-1}$, where u_t is a white noise process with variance σ^2 .
- [1] How many non-stationary solutions does this equation have?
 - [2] How many stationary solutions of $MA(\infty)$ form with respect to (u_t) does this equation have?
 - [2] Plot all roots of the AR lag polynomial on the complex plane and indicate the unit circle.
 - [3] Rewrite all stationary solutions in $AR(\infty)$ form with respect to (u_t) , if possible.
 - [2] Find $\mathbb{E}(y_t)$ for all stationary solutions.
5. Consider the stationary solution of the equation

$$y_t = 0.3y_{t-1} + 0.4y_{t-2} + u_t,$$

where (u_t) is a white noise process.

- [3] How many stationary solutions does this equation have?
 - [1] Check the invertibility condition for this equation.
 - [3] Find the values of ACF $\rho(1), \rho(2), \rho(3)$ for the stationary solution.
 - [3] Find the values of PACF $\phi(1), \phi(2), \phi(3)$ for the stationary solution.
6. For each of the following equations:
- Check whether a stationary solution exists.
 - Check whether the process is invertible.
 - Identify the process as $ARIMA(p, d, q)$ with the smallest possible d ($d = 0$ for stationary process).
 - If the process is non-stationary, suggest a transformation to make it stationary.
 - If the AR and MA polynomials have common roots, state them.
- [2] $y_t = 0.5y_{t-1} + 0.5y_{t-2} + u_t - 0.5u_{t-1}$
 - [2] $y_t = 1.5y_{t-1} - 0.5y_{t-2} + u_t - 0.2u_{t-1}$
 - [3] $y_t = y_{t-1} - 0.25y_{t-2} + u_t - u_{t-1} + 0.25u_{t-2}$
 - [3] $(1 - L)^2 y_t = (1 - 0.8L)u_t$